SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Let Skills Evolve Collectively with Agentic Evolver

arxiv.org/abs/2604.08377

ML Researchers & AI Systems Engineers | Technical Presentation

Agent Skills Remain Static After Deployment

Static Installation

Skills are manually installed from a hub and never updated from runtime experience.

No Cross-User Learning

Solutions discovered by one user do not propagate to others — each rediscovers independently.

Recurring Failures

The same failure-recovery patterns are observed repeatedly, yet the system does not improve.

Existing Approaches Fall Short

Memory-based

Store past trajectories but tied to specific instances; hard to generalize

Skill-based

Compress experience into instructions but treat library as static — no evolution through usage

SkillClaw closes the loop

Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution

1 EVIDENCE

Aggregate cross-user session trajectories by referenced skill

2 ATTRIBUTION

Diagnose root causes of failures & successes across user contexts

3 EVOLUTION

Propose skill mutations via open-ended agentic reasoning

4 DISTRIBUTION

Verify updates, merge & synchronize to all agents

← Continuous closed loop — evolved skills feed back into multi-user interaction →

Multi-User Agent Ecosystems

OpenClaw • CoPaw • IronClaw • PicoClaw • ZeroClaw • NanoClaw • NemoClaw

Agentic Evolver: Evidence → Attribution → Evolution

1 EVIDENCE

Analyze trajectories to identify behavioral patterns across users

- Parse structured session logs
- Group by referenced skill
- Identify recurring success/failure patterns

2 ATTRIBUTION

Diagnose root causes of failures and successes

- Map failures to specific skill instructions
- Separate generalizable issues from one-off edge cases
- Cross-user correlation reveals true gaps

3 EVOLUTION

Propose skill mutations — refine existing or create new

- Generate candidate skill updates
- Open-ended reasoning, not predefined rules
- Autonomous agent driving the process

Key distinction: open-ended agent reasoning over interaction evidence — not predefined update rules

Cross-User Trajectory Aggregation Exposes Skill Boundaries

Different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary — revealing both where it works and where it breaks.

User A

Context α

Success patterns

User B

Context β

Failure modes

User C

Context γ

Edge cases

User D

Context δ

Recovery paths

Aggregated Skill Evidence Base

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix. Cross-user aggregation is essential.

Compatible: OpenClaw • CoPaw • IronClaw • PicoClaw • ZeroClaw • NanoClaw • NemoClaw

Case Study: Multimodal Creative Pipeline — 1 of 6 Accepted

Day	Candidate Skill	Result	Rationale / Next Action
1	validate-tmp-workspace-inputs	✓ Accept	Check /tmp_workspace inputs & env setup → Upgrade to new best pool
2	multimodal-input-validation-and-setup	✗ Reject	Multimodal input validation & output env init → Keep current best pool
3	multimodal-creative-task-pipeline	✗ Reject	Extract from PDF/video/image pipeline → Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	✗ Reject	Added image classification, visual generation, structured validation → Keep current
5	pipeline (impr.) + validate-required-input-files	✗ Reject	Creative pipeline + per-file fail-fast validation → Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	✗ Reject	Extended PDF-to-poster / document-to-visual paths → Not admitted to next cycle

Conservative verification: only 1 of 6 candidates accepted over 6 days — prevents regressions

Case Study: Git Push Auth-Fallback — 3 of 6 Accepted

Day	Candidate Skill	Result	Rationale / Next Action
1	git-push-with-auth-fallback	✓ Accept	Safe fallback instead of blocking on push failure → Add to Safety best pool
2	git-push-with-auth-fallback	✓ Accept	Unified patch/bundle filenames, reduced inconsistency → Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	✓ Accept	Auth-alternative paths & secrets audit; fixed subshell pitfalls → Keep current
4	(same pool retest)	✓ Accept	Validator confirmed current pool as best → Continue same best pool
5	git-push-with-auth-fallback	✗ Reject	Added 'push hang treated as auth failure' — no improvement → Keep current
6	git-push-with-auth-fallback	✗ Reject	Added identity config & filename consistency — no improvement → Not admitted

Iterative refinement succeeds when evidence supports it: 3 accepted updates before plateauing

Three Properties That Distinguish SkillClaw

01 Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. Cross-user aggregation unlocks signals no single user can generate.

02 Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. The system improves as it is used.

03 Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning over interaction evidence — not predefined update rules. An autonomous agent drives evolution.

Implication: Agent systems can shift from static skill libraries to living, collectively-maintained knowledge ecosystems that improve with every user interaction.